

CELLCHECK · DESIGN DECISION RECORD

CellCheck recommendations accepted

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DOCUMENT CONTROL

Document ID	CCK-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cellcheck		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Items 10, 12, 14 and 16 decided by Amish on 2026-10-02 as recommended in CCK-DEC-001
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all recommendations and what changed in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted by Amish. Every item with a recommendation in docs/REVIEW.md and CCK-DDR-001 is Decided by Amish, 2026-09-25: go with recommendation. Items without a recommendation (10, 12, 14 and 16) were given recommendations in the design decisions register (CCK-DEC-001, items 3 to 6) and decided by Amish on 2026-10-02: "i approve your recommendations for all 555 open decisions."

■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Before that, the CellCheck recommendations had been adopted for TRL 3 work only, open for his review (CCK-DDR-001 v0.1). This record lists each newly decided item, what changed in the repo because of it, and what stays open. Where a recommendation offered several options, the recommended option is the decision. The portfolio stays at TRL 3; work that needs a build, a test, firmware beyond a sketch or purchasing is recorded as decided but on hold, because TRL 4 is on hold by Amish's instruction.

■ Decision

Table 1. Items decided on 2026-09-25 and the resulting changes.

# (DDR-001)	Decision	What changed in the repo
1	Keep SwapCell in the pitch and plan a second-life SwapCell variant (option a)	Wording in CCK-PRB-001 v0.4, CCK-PRC-001 v0.4 and the README changed from "adopted, open for review" to decided. Pitch and problem lines unchanged (no rewording was recommended). Agreement with SwapCell and PowerBox is a cross-repo action
2	Charge and discharge attended only; the rest stage may be left	Wording only; R14 as redefined in CCK-REQ-001 v0.3 stands
3	Hardware watchdog and per-channel undervoltage comparators	Wording only; BOM line 18 and model part 18 were already in place
4	Budget raised to \$175 (option a)	<code>project.yaml</code> <code>budget_usd</code> 160 to 175; R12 target in CCK-REQ-001 v0.4 \$160 to \$175; R12 status Not met (\$4.00 over) to Met (\$11.00 under); CCK-CAL-001 v0.2 section 11 and <code>sizing.py</code> ; README budget line; BOM notes
5	Grading thresholds, 14-day rest with 50 mV limit and ± 1 % group tolerance as starting values	Wording only. Tuning with the first 100 real cells is TRL 4 work: decided, on hold
6	TP5100, INA226 and ESP32-S3 class modules	Wording only. Purchasing is TRL 4: on hold
7	Eight channels with a linear load	Wording only
8	LFP profile later	Wording only
9	Per-cell CSV mapped to ReflowEconomy's passport v0.2	Wording only
11	DC four-wire resistance and a 14-day rest	Moved from "Proposed, awaiting Amish" to decided in CCK-PRC-001 v0.4; CCK-CAL-001 already used them
13	Firmware and build rules from CCK-CAL-001: same-channel, cohort-median self-discharge reading; resistance pulse only within 1 K of the bench; per-channel calibration; fan-fault stop; detached-thermistor plausibility check	<code>sizing.py</code> now uses a 1 K reinsertion difference (was 2 K): R4 repeatability ± 2.99 to ± 2.09 m Ω worst case (± 2.02 to ± 1.28 m Ω RSS), R4 At risk to Met. R6 is now evaluated in the same channel (± 10.6 to ± 2.3 mV) but stays at risk because the spread of relaxation is not bounded. The rules are written into CCK-PRC-001 v0.4 steps 3 and 6 and the safety section. Firmware, calibration and tests are TRL 4 work: decided, on hold
15	Raise the two passport schema gaps with ReflowEconomy	Listed under cross-repo actions in <code>docs/REVIEW.md</code> ; no other repo edited

No part was added or resized, so `cad/src/model.py` , the STEP and STL files and the general arrangement CCK-DWG-001 are unchanged in geometry and notes; the drawing stays at Rev P1 and was only re-rendered.

Table 2. Requirement status after this record (CCK-CAL-001 v0.2).

Status	Requirements
Not met	None (R12 was not met before this record)
At risk	R5, R6, R11
Not verifiable at TRL 3	R10
Met on paper	R1, R2, R3, R4, R7, R8, R9, R12, R13, R14, R15, R16, R17 (13 of 17; was 11)

■ Items left open by this record

Table 3. Items that had no recommendation here, decided by Amish on 2026-10-02 as recommended in CCK-DEC-001.

# (DDR-001)	Item	Decision of 2026-10-02
10	First trial partner: repair café, e-bike repair shop or battery collection point (no preference stated)	A local Repair Café group that passes laptop batteries to a collection point, with that point as the source of trial cells; the first candidate to approach, not yet agreed (CCK-DEC-001, item 3)
12	Scheduling rule for the attended day: finish current stages the same day, or pause overnight with the unit off	A charge or discharge starts only if it finishes the same day (CCK-DEC-001, item 4)
14	Responses to R11 (mass) and R5 (throughput)	Accept 4.91 kg with the 1.5 mm tray and weigh at TRL 4; throughput by the same-day rule (CCK-DEC-001, item 5)
16	Grade A resistance limit per cell model (needs named data sheets)	60 mΩ default; per-model limits from measured DC resistance for the commonest models only (CCK-DEC-001, item 6)

■ Consequences

- CellCheck has no requirement failed outright on paper. R5, R6 and R11 remain at risk and R10 cannot be verified before a physical test.
- Cross-repo actions (SwapCell and PowerBox for the second-life pack, ReflowEconomy for the passport gaps) are listed in `docs/REVIEW.md` ; no other repo was edited.
- `trl` and `trl_target` stay at 3. TRL 4 is on hold by Amish's instruction.